

Distributed sterile-injectable manufacturing under real-world constraints

Executive summary. Rikhin Kavuru. Full memo and code: `reports/memo/memo.pdf, docs/, src/`.

PRELIMINARY AND ILLUSTRATIVE. NOT FOR EXTERNAL CLAIMS. 87 of 91 model parameters are tier-5 placeholders with no operational, official, peer-reviewed or elicited source. Zero of a target 25 to 30 interviews and zero of three required independent reviews are complete. All sixteen regulatory gates are UNCERTAIN, so eligibility is NO_CONCLUSION for every architecture. Nothing here claims any system is FDA-compliant, clinically deployable, or ready to manufacture drugs. Every number is the behaviour of a model under those inputs.

The question

Under what product, demand, regulatory and operating conditions could regional manufacturing reduce sterile-injectable shortages more effectively than safety stock, dual sourcing, or additional centralized capacity? Two non-controlled small-molecule presentations were carried through: sodium bicarbonate 8.4% in a 50 mL vial, in prolonged shortage, and norepinephrine bitartrate 1 mg/mL in a 4 mL vial, resolved.

Method, in one paragraph

A protocol frozen before any run fixes the service target: mean fill rate ≥ 0.99 in at least 90% of simulated years. A daily discrete-event engine simulates twenty supply architectures over five years on common random numbers, so every architecture meets the same disruptions in the same years. Each is optimized to that one target, then re-evaluated at 100 paired runs. An ablation harness over 22 named failure mechanisms attributes the result and, in practice, detects defects: a factor whose sign comes out wrong is usually a bug. Twenty-five were found that way.

What the model says

- **A capacity-short network cannot be stocked out of the deficit.** On the short product, a full year of safety stock with daily base-stock review leaves the status quo at mean fill 0.869 and a tail probability of zero. At twice base demand, none of the twenty architectures meets the target anywhere. That region argues for capacity, not for *regional* capacity.
- **Owned regional nodes lose the years before they exist.** On the short product the distributed design needs commissioning within 741 days, supplier disruption at or below 0.127 against an assumed base of 0.200, shelf life at or above 17.6 months and changeover within 4.5 days. It therefore sits outside its own feasible region on the supplier axis at the base assumption, at the highest cost in the battery: 32.0 against 19.4 million USD per year for the cheapest cell that meets the target.
- **Adding a validated release layer to those nodes buys nothing.** It returns mean fill identical to four decimals while costing 1.0 to 1.7 million USD per year more. The reasons are a model defect (MD-18) and a regulatory gate held UNCERTAIN (G15), neither an argument for the layer.
- **Regional manufacturing is not eliminated, but the honest form is narrow.** On the product that is *not* capacity-short, the distributed design meets the target at a tail probability of 0.940, at about 9% above rotating contracted campaigns and 32% above the cheapest feasible cell. It clears the bar where the incumbent plant has slack, and not where shortages actually arise.
- **Two architectures meet the target on both products, and neither is a plant the sponsor builds.** Rotating prequalified campaigns across three already-registered third-party fill lines (S17), and dual finished-dose sourcing with an explicit key-starting-material tier (S9).

The conditional answer

On these inputs, the strongest intervention is contracted registered capacity plus positioned finished-goods stock, bought rather than built. The binding constraint is a contract, not a machine. Of 40 re-evaluated cells, 14 meet the target and 16 sit within one standard error of it, so two fifths of the verdicts cannot be resolved at this sample size.

What would overturn this, and what it would take

The three inputs that most change the answer are the regional replenishment rule, the rate at which one event removes several nominally independent sites, and the real backorder window. All three are currently placeholders the author wrote. None can be settled by more computation; each needs a practitioner. The study also retracted three of its own headline conclusions after correcting its own model, and one earlier failure of the distributed design proved to be an artifact of an unequal search space rather than a property of distribution.